

Chandrika Sethumadhavan

Female (She/her) — chandrikasethu06@gmail.com — LinkedIn — Github — Freiburg, Germany

DO I FULFILL YOUR PROFILE?

- 5+ projects in embedded systems, machine learning, and computer vision. Published in SSRN 2025 and Springer 2023.
- Professional experience at Fraunhofer ISE and IMTEK in sensor interfacing and ML-based defect detection.
- Applied deep learning techniques (YOLOv8, Mask R-CNN, segmentation, enhancement) for industrial applications.
- Experience deploying ML models using Flask, AWS CI/CD, and cloud-based apps on projects.

TECHNICAL SKILLS

Programming: Python, C, C++, SQL

Image Processing & Machine Learning: *Libraries:* OpenCV, YOLOv8, TensorFlow (+Lite), Edge Impulse, PyTorch, NumPy, SciPy, scikit-learn, Optuna *Concepts:* TinyML, MLOps, AutoML, Model Deployment, Experiment Tracking, Hyperparameter Tuning, Data Preprocessing, Model Evaluation, Carbon-Aware ML

Embedded Systems: *Hardware:* Raspberry Pi, MSP430, ESP32 *Skills:* Microcontroller Programming, Sensor Integration, PCB Design, Testing & Debugging, Firmware Flashing, Keil uVision

IoT Data & Visualization: MQTT Protocol (TTN integration), InfluxDB (Cloud, Time-Series DB), Grafana (Dashboard Config, Flux Queries)

Deployment & Tools: *DevOps/Platforms:* AWS (Elastic Beanstalk, EC2), Docker, GitHub, GitLab, CI/CD, FastAPI *Environments:* Flask, VS Code

EDUCATION

University of Freiburg	Germany
<i>M.Sc. in Embedded Systems Engineering</i>	2023 – present
GPA: 1.9	

Anna University	India
<i>B.E. in Electrical, Electronics Engineering</i>	2019 – 2023
GPA: 1.4	

WORK EXPERIENCE

Master Thesis at IMTEK, Uni Freiburg	Germany
<i>Master's Thesis Researcher</i>	Ongoing

- Built a physics-informed LSTM/GRU transfer learning pipeline trained on CaO₂ oxygen-release dynamics and fine-tuned to predict ANT-EPO pressure-time profiles under sparse experimental data.
- Hybridized first-order kinetics, gas–liquid partitioning, and ML residual correction to model sealed-chamber pressure evolution for microactuator design. mapped feasibility regions to facilitate design of microactuator materials for biomedical applications.
- Applied multi-objective optimization (NSGA-II) to extract constraint-satisfying operating windows; benchmarked physics-only, data-only, and transfer learning variants on RMSE, MAE, and R².

Work Student at ECOSENSE, IMTEK, Uni Freiburg	Germany
<i>Working Student – Embedded Systems</i>	Sep 2024 – Nov 2025

- Designed, assembled, and tested PCBs using **Eagle** for environmental sensors; firmware flashing and debugging with **Keil uVision**.

- Developed Python-based MQTT logging for real-time sensor data from The Things Network (TTN) with automatic CSV export and interval-check logic.
- Integrated InfluxDB Cloud and configured Grafana dashboards for live monitoring of sensor values.

Work Student at Fraunhofer ISE, Freiburg

Germany

Working Student – Image Processing & ML

Dec 2023 – Jun 2024

- Optimized YOLOv8 inference for industrial production-line vision systems, integrating feature extraction, edge detection, and thresholding to enable live defect detection on solar panels with prediction accuracy up to **99.4%**.
- Utilized **OpenCV, PIL, NumPy, and SciPy** for precise statistical analysis of mesh screen irregularities.
- Integrated hardware-software pipeline with industrial 3D Measuring Laser Microscope (LEXT) for automated quality inspection and optimization of solar cell finger-line printing.

KEY PROJECTS

(Coursework) Zero-Cost Proxy AutoML: Carbon-Aware Neural Architecture Search Github

- Built a resource-aware AutoML pipeline combining zero-cost proxies and carbon-aware techniques for efficient neural network model selection on vision datasets.
- Integrated CodeCarbon for carbon footprint tracking during hyperparameter optimization using Optuna, minimizing environmental impact.
- Achieved **86.26%** accuracy and **0.796 F1-score** on skin cancer classification dataset using PyTorch and TorchVision.
- Evaluated across multiple open-source datasets (flowers, emotions, fashion, skin cancer) with accuracy ranging from 64.6% to 97.0%.

(Coursework Extension) Deployment-Aware AutoML – ONNX Inference Optimisation Github

- Extended the AutoML pipeline with a deployment stage: automatic ONNX export, dynamic INT8 quantisation, and structured multi-backend inference benchmarking across PyTorch CPU, ORT CPU, and ORT CUDA.
- Achieved **1.87×** CPU speedup and **8.54×** GPU speedup with no accuracy loss via ONNX Runtime graph-level optimisations (operator fusion, memory layout improvements).
- Reduced model size by **74.8%** (54.71 MB to 13.78 MB) through dynamic INT8 quantisation, enabling bandwidth-constrained edge deployment.

Surgical Documentation Assistant – Black Forest Hackathon (SxH track) 2025 Github

- Integrated OpenAI API for automated surgical report generation from medical image analysis; pipeline processes surgeon-selected frames and produces structured clinical documentation in real time.
- Built PyTorch-based ML pipeline using EndoViT (medical vision transformer) for surgical scene segmentation and phase classification from video.
- Developed React/TypeScript tablet-optimized frontend with swipe-based image selection, AI quality filtering, and report export – shipped end-to-end within the hackathon timeframe.

Meals or Miles: Recipe Carbon Footprint Analyzer – Black Forest Hackathon, Offenburg 2025 Github

- Built a recipe analysis tool that calculates meal carbon emissions and suggests sustainable ingredient alternatives using vector embeddings for multi-language ingredient matching.
- Implemented multi-objective optimization to identify key emitters (>15% of total carbon) and generate nutritionally-balanced replacements, reducing carbon footprint while maintaining similar protein, carbs, and fat.
- Flask-based backend with real-time comparison visualization; zero API calls and minimal computational overhead.

Microcontroller Programming with MSP430

Github

- Developed embedded systems using the MSP430 microcontroller, including LED toggling, shift registers, ADCs, interrupt handling, and timer configuration.
- Implemented Watchdog Timers (WDT) for system reset in case of deadlock, with task management through preprocessor directives for real-time operations.

(Coursework) “Prediflex” Smart Insole – Microsystems Design Lab

Github

- Virtual design and analytical modeling of an energy-harvesting microcontroller system powered by passive piezoelectric sensors that convert biomechanical energy into usable electrical power.
- The harvested energy supports both the microcontroller and wireless communication modules, conforming to international standards.
- System architecture emphasizes ultra-low-power operation, sustainable materials, and seamless integration for real-time pediatric gait analysis with predictive assessment of abnormalities.

CERTIFICATIONS

Tiny Machine Learning (TinyML) – HarvardX (edX, Audited)

Harvard University

Deploying ML models on microcontrollers using TensorFlow Lite Micro, model optimization via quantization and pruning, and TinyML applications for keyword spotting, anomaly detection, and gesture recognition on ultra-low-power edge devices (Arduino, ESP32).

AWS Certified Machine Learning Engineer – Associate (MLA-C01)

Amazon Web Services

End-to-end ML pipelines on AWS: SageMaker (training, endpoints, Pipelines), data engineering with Glue and S3, model monitoring, drift detection, and MLOps best practices.

PUBLICATIONS

Kinetic Modeling and Quantification of Oxygen Release from Anthracene Endoperoxides under Confined Conditions SSRN, DOI: 10.2139/ssrn.6504441, 2025.

Autonomous Delivery Vehicle Using Raspberry Pi and Computer Vision – Springer, Proceedings of the International Conference on Pervasive Computing and Communication for Data Analytics (PCCDA’23), 2023.

LANGUAGES

English – C2 — German – B1